

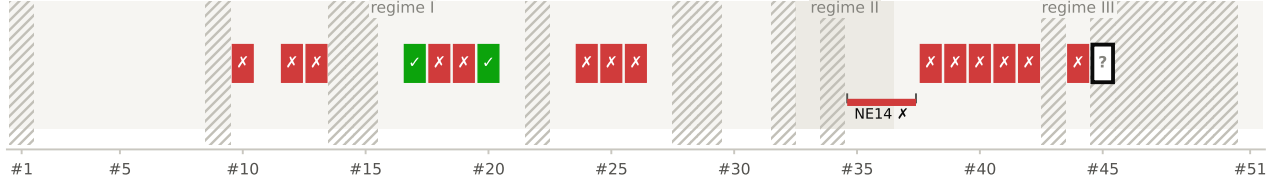
H02: Inferred couplings reflect real influence

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 0.5/5 Usefulness 1.5/5

Bottom line. Activity-timing Ising couplings do not measure influence: they sit at the null floor, miss installed leaders, and the collective term is the daily schedule.

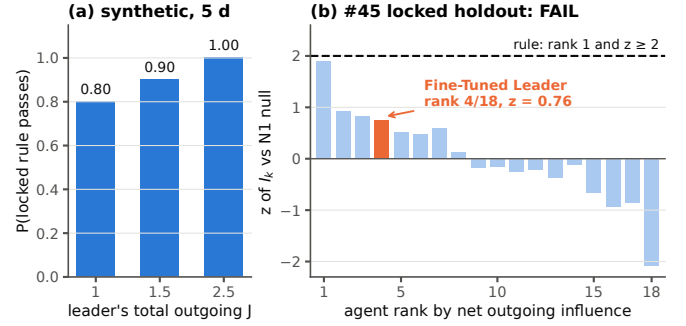
level: hypothesis · pipeline checks 99% · updated 2026-10-04



Goal periods tested (2 supported, 15 failed). ✓ supported ✗ failed ✗ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

The spins are the 1-min activity states of the agents. A directed kinetic-Ising fit gives couplings in 21 non-holdout 5-day chunks (#10–#44), with block fields and surrogate nulls. In synthetic villages the pipeline recovers edges (AUC 0.95) and finds a leader with outgoing $J \geq 1.5$ in 90–100% of runs. On real data the pairwise couplings sit at the null floor in both goal modes. The fit misses #26's elected leader. **The locked holdout #45 fails:** the Fine-Tuned Leader ranks 4/18 at $z = 0.76$. **Round 1b** corrects the minute grid, which dropped half of all events, and trims day edges in the nulls. The pairwise couplings stay null. The round-1 collective coupling in regime III is the operator's day edges: it is significant in 2/8 regime-III chunks, against 7/13 in regime I. The fit also misses #12's debate teams and #44's installed leader. Post hoc, talk-spin couplings transfer across days in regime-III shared-objective weeks.



Observables. (a) The pass rate of the locked rule in synthetic villages sampled like #45 (18 agents, 5 days), against the planted leader's total outgoing coupling. The rule finds a leader of strength 1.5 or more in almost every run. (b) The real #45 holdout: z of net outgoing influence for every agent. The installed Fine-Tuned Leader ranks 4th at $z = 0.76$. No agent reaches $z = 2$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An inverse-Ising fit to activity timing does not identify who leads an agent swarm. Per-pair couplings from activity timing are noise. They miss a leader that the operators installed twice (#45 and #44), and they miss a debate's team structure. The collective co-activation that looked like coupling is mostly the operator's daily start and stop. Before you compute any synchrony statistic, trim each day to the all-present window. Without the trim almost every regime-III chunk looks coupled; with it, 2/8 do. To find which agent steers a swarm, measure who talks after whom. That pairwise signal survives, consistent with H50's read-out-gated talk coupling.

PHYSICAL MODEL

Model 02 (kinetic Ising) is primary; model 01 (equilibrium Ising) is the rival. The spin $s_i(t) = \pm 1$ says whether agent i acts or talks in 1-min bin t . h_i is a bias. The field $\delta_i(b)$ per (day, 30-min block) absorbs the schedule and kickoffs. The coupling J_{ij} is the lagged influence of j on i ; σ is the logistic function. I_k is the net outgoing influence. In mean field, βJ_0 is the uniform coupling, m the mean spin and q the mean spin variance. VR is the variance of $\sum_i s_i$ over its independent value.

$$P(s_i(t+1) = +1 | \mathbf{s}(t)) = \sigma(2H_i(t))$$

$$H_i(t) = h_i + \delta_i(b) + \sum_j J_{ij} s_j(t)$$

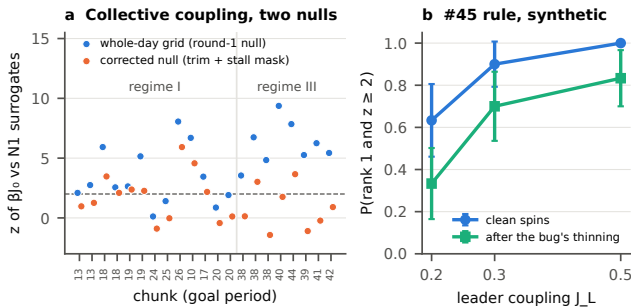
$$I_k = \sum_{j \neq k} (J_{jk} - J_{kj})$$

$$N \text{Var}(m) = q/(1 - \beta J_0 q) \Rightarrow \beta J_0 = (1 - 1/\text{VR})/q$$

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b, Verdict corrected null)
SP1–SP5 synthetic recovery	AUC 0.95; leader top-1 1.00 at 4 of 5 $J_L \geq 0.3$; FPR 5.8–7.6%
RP1 mode C $\geq 1.5 \times$ mode I, above floor	6–8% → 5–11%; C/I 0.63, $\times$ 1.40
RP4 pairwise beats M1 on held-out days	2/21 → 3/21 chunks vs N1 $\times$
RP6 same-lab enrichment ≥ 1.3 ; RP7 #26 leader top 3	1.16 → 0.85; 8/10 → 10/10 $\times$
MF-P1 $\beta J_{0q} < 0.5$ everywhere	max 0.36 → 0.34 $\checkmark$
MF-P2 III-C significant, mode I null	III-C 5/5 → 2/5; regime I partial 7/13
MF-P3 forward $P(K)$ (unfitted)	CW better 5/13 → 6/9 partial
#45 holdout (rank 1, $z \geq 2$)	4/18, $z = 0.76$ (buggy table; $\times$ not re-run)
Native: #12 teams; 44b leader; NE14	$z = -0.32$; rank 1/5, $z = \times \times \times$ 0.94; rise → -0.08



Second figure. (a) z of the collective coupling βJ_0 per chunk on the corrected grid. Against the round-1 whole-day null almost every chunk looks coupled. Against the corrected null (day edges trimmed, stalls masked) regime III collapses, except #44. (b) Synthetic: the event-drop bug cuts the #45 rule's power from 0.90 to 0.70 at $J_L = 0.3$. It rarely turns a strong leader into $z = 0.76$.

GOAL PERIODS IN DETAIL

G45 ? 1b: not re-run (locked holdout; the executed run read... (**holdout**)
G10 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G12 $\times$ failed
G13 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G17 $\checkmark$ 1b: supported corrected data and DQ8 null: significant-c...
G18 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G19 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G20 $\checkmark$ 1b: supported corrected data and DQ8 null: significant-c...
G24 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
G25 $\times$ 1b: failed corrected data and DQ8 null: significant-c...
+8 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	0
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The corrected null still leaks under start-up bursts: its DQ8 size is 0.51 for day-edge drives after trimming. H38's agent-state conditioning agrees with it here. The trim keeps 33–99% of minutes, and only 33% in #44. The #45 run used the buggy table. The thinning check draws day retention from non-holdout regime-III days, not from #45. The native tests are small: 295 debate-minutes for #12, and two days and five agents for 44b. The talk-spin result is post hoc. The synthetic harness does not use the corrected calibration (active 0.67, J_{self} 0.49).

ROUND 2

Measure influence in messages: H02-R1 uses context-ledger visibility. Pre-register the talk-spin coupling in regime-III shared-objective weeks before any confirmatory use. Vivian decides whether to re-run the #45 rule on the fixed table.

- **Essence.** Who changes whose mind?
- **H02-R1** Influence is the change in j 's next decision when i 's message is in j 's context (context ledger), compared with matched turns where it is not.
- **H02-R2** The #45 leader influenced plans, not activity: assignment-acceptance events (Jev-labeled) show its authority.
- **H02-R3** Influence follows authority markers (operator endorsement, titles, a leader role), not model family.

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